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An ensemble learning model for predicting the intention to quit among employees using classification algorithms

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ABSTRACT

Employees are often more likely to use social media for job searching, which sometimes causes withdrawal behaviour. This study proposes an ensemble learning model for predicting the intention to quit (IQ) based on selected features, such as job Involvement (JI), organizational commitment (OC), activities on professional networking sites (APNS), and updating profiles on job portals (PJP). The Receiver Operator Curve (ROC) examines the model's accuracy. We show the best relationship to predict intention to quit is between activities on professional networking sites and updating one's profiles on job portals on social media. Seven classification algorithms of Gradient Boosting, Random Forest, K-Nearest Neighbour, Logistic Regression, Neural Network, Support Vector Machine, and Naïve Bayes are used to build the classification model. In addition, four combinations of the above-mentioned methods are used to construct an ensemble learning classification model. The performance comparison indicates that the combination of Gradient Boosting, Logistic Regression, and K-Nearest Neighbour produced the best ensemble learning model for predicting the intention to quit. The study's contribution incorporates the stimulus organism response theory to predict the intention to quit through information gain, emphasizing social media features. Based on these features, the classification tool is utilized to identify between those who intend to resign and those who do not.

1. Introduction

Employee turnover varies by industry but is much higher in the IT sector than in government companies [1]. According to Mohammad and Jahangir [2], turnover intention is the desire of an employee to leave the organization. According to National Association of Software and Service Companies (NASSCOM), which is the premier trade body and chamber of commerce of the tech industry in India, data from 2019, the global workforce turnover rate ranges from 18 to 20%. In this case, the organization must lower the risk of turnover [3]. Employee turnover has an impact on the cost of rehiring, financial losses, the loss of a highly skilled workforce, executive competence gaps, knowledge acquisition, knowledge transfer channels, high hiring and training expenses, and organizational performance in terms of productivity and revenue. As a result, the company constantly struggles to predict employee intentions to quit (IQ) [3–6]. Thus, it is crucial to develop a solid model to predict employee quitting intentions so that the human resources manager (HR) will subsequently be able to take the initiative to retain the most competent employees in the organization.

Every industry has seen a transformation as a result of recent improvements in artificial intelligence, which have also produced creative algorithms that outperform humans in a range of learning tasks or are on par with them. Predictive models in Human Resource Analytics (HR Analytics) based on machine learning (ML) have also been employed for a wide range of commercial challenges. Employee turnover has been extensively predicted using machine learning algorithms that utilize ensemble techniques [7]. To predict employee turnover, various classifiers such as supervised machine learning algorithms, Support Vector Machine (SVM), Naive Bayes, K Nearest Neighbour (KNN) classifiers were used to look at job characteristics, performance, age, level of education, geographical characteristics, salary, working conditions, job satisfaction, supervision, advancement, recognition, growth potential, burnout, time spent, and work hours per month [8,9]. Furthermore, Job satisfaction, organizational commitment, job involvement [10], and enterprise social media (utilitarian use, hedonic use, and social use) are the indicators that predict the employee's intention to quit [11]. Also, the employee behavioural assessments are connected to the turnover decision based on feature filtration [12]. Chung et al. [13]

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consider personal attributes and Job-related variables for modelling the machine learning-based prediction model for employee attrition. Thus, the previous research concentrated solely on demographic and behavioural factors in predicting employee intention to quit using machine learning techniques [12]. Along with the traditional methods, machine learning methods are also used for predicting the intention to quit. Alduayj and Rajpoot [14] applied three distinct methods, namely, KNN, Random Forest and SVM for predicting the intention to quit. The performance comparison between the three methods indicates that Random Forest is superior to the other two methods. Fallucchi et al. [15] use Naïve Bayes, logistic regression, KNN, Decision tree, Random Forest and SVM for predicting the intention to quit. The performance comparison shows that the Naïve Bayes method produces better result in prediction. Najafi-Zangeneh et al. [16] show that Naive Bayes is the best method for predicting intention to quit in comparison to K nearest neighbour, decision tree, Random Forest and Logistic Regression. All these above articles use the IBM HR analytics data, which contain only demographic information. So, there is a need to conduct the study in different way. As a result of that the same problem is studied by using different features, which are identified in the literature review section. Additionally, there are very few studies using the ensemble learning classification model to predict Intention to Quit (IQ) by examining the relationships between the predictors or features. The association between the predictors or features, which is a stimulus to predict IQ as a response, has not been demonstrated by prior studies. This methodological gap is closed by placing an emphasis on the ensemble learning classification model and developing a relationship between the predictors as a tool for IQ prediction.

Social media-based predictions of employee turnover are still in their infancy, though, given how rapidly society is being disrupted by social media. Advanced analytics techniques can be utilized with social media data to build models that predict employee's attrition. Therefore, it is crucial to develop a reliable and predictive model that estimates the likelihood that an employee would quit the organization by examining existing data and identifying trends using HR Analytics [17]. It is discovered that updates on professional advancement impact job search behaviour using the theories of social contagion on SNSs and social comparison theory [18]. Also, workers who frequently update their social media profiles may be more likely to consider leaving their jobs [19]. Early identification of these signals allows organizations to take proactive measures to address problems and perhaps retain valued talent. Hence, it is crucial to distinguish between employees who intend to quit and those who do not, based on the predictors and thereby building up a robust model. Thus, the study intends to raise research questions such as

RQ1: What ensemble learning techniques are suitable for predicting IQ using the selected features or predictors?

RQ2: What is the optimal combination of ensemble learning algorithms for achieving the highest IQ prediction performance?

RQ3: What are the key statistical methods to assess the relationship between the selected features or predictors?

RQ4: Which predictors exhibit the strongest relationship with IQ prediction?

RQ5: How can the Stimulus–Organism–Response (SOR) theory be integrated into IQ prediction?

The theoretical and methodological gaps between earlier research and the strategy has been employed in the current study to provide a driving force for this research work in this area of HR Analytics. In this study, an ensemble learning classification model is applied to predict an individual's intention to quit their job based on a number of features or variables, such as job involvement (JI), organizational commitment (OC), job satisfaction (JS), activity on professional networking sites (APNS), and updating profile information on a job portal (PJP), which were not considered in previously published literature for building machine learning based models. The Receiver Operator Curve (ROC) is used to assess the accuracy of the model for representing the best relationship between the selected variables or features in predicting intention to quit (IQ). Gradient Boosting, Random Forest, K-Nearest Neighbour (KNN), Logistic Regression, Neural Network, Support Vector Machine (SVM), and Naive Bayes are seven classification methods used to build the classification model and then the best combination of multiple classification algorithms has been combined through ensemble learning technique. With a focus on social media variables, the study integrates the stimulus organism response theory [20] to forecast the intention to quit through information gain. Results highlight the efficiency with which the proposed framework is employed to create a reliable and accurate predictive model.

The current research provides three contributions to the field of research in HR Analytics. The study focused on social media predictors for employee IQ, such as activities on professional networking sites (APNS) and updating a profile in the job portal (PJP), which are rarely used in the previous literature. Second, determining whether the employees intend to quit by applying a classification tool that uses ensemble approaches. Finally, the application of the stimulus-organism response theory to better understand the employee's behaviours resulting in the intention to quit or not intending to quit

The organization of the article starts with an introduction to the research problem. The related studies are reviewed in Section 2. The methodology adopted in this research article the and data collection process have been described in Section 3. The experimental findings are reported in Section 4, together with model analysis and insights. In Section 5, the conclusion has been provided and finally, the work's restrictions and its future directions are discussed in Section 6.

2. Literature review

2.1. Intention to Quit (IQ)

Rarely do employees start and finish their careers at the same company. Such a circumstance is not frequently encountered. The literature identifies the intention to quit when an employee desires to leave their job [2]. Employees actively seek new employment, which raises the probability of leaving by choice [21–23]. Employees show their covert job search behaviour when they explore their career options, take part in more training courses, and use job search clarity [24]. Furthermore, with the introduction of numerous social networking platforms, employees can now search for jobs and build their networks during their free time, resulting in higher employee turnover [11]. This behaviour among the employees in the workplace has a direct impact on the organization's overall success [25]. Furthermore, previous research investigated the role of organizational commitment, job satisfaction, and job involvement in predicting an employee's intention to quit (IQ). Highly dedicated personnel are content with their jobs and are less likely to quit [26,27]. When an employee's skills in their area improve, exposure to career growth prompts them to look for new employment opportunities. With the advancement of technology, social media has become an important tool for growing networks and creating a job-search environment. Professional networking platforms on social media, such as "Linked In", give a platform for building a network for job hunting, hence improving employability (Habets et al., 2021). Furthermore, regular use of social media networking sites suggests a higher staff turnover rate [11]. Therefore, the use of social media for non-work-related purposes increases turnover. Also, according to our prior research, an unhappy or unsatisfied employee with less involvement and frequent activities on professional networking sites is more likely to intend to leave the organization. Therefore, employees' covert behaviour towards their intention to quit (IQ) in a non-work-related situation is recognized through social media usage [19].

2.1.1. Job Involvement (JI), organizational Commitment (OC), and Job Satisfaction (JS) towards IQ

Employees desire to quit or turnover is seen as a major issue for any firm in the age of globalization. The antecedents of employees' intentions to quit have been the subject of debate for decades. The broad characteristics discovered in predicting an employee's intention to quit include job involvement, organizational commitment, and job satisfaction [28]. Lodahl and Kejner's [29] definition of job involvement as the degree of psychological identification with one's work. The findings of past research suggest that highly job involved employees rarely think of quitting the organization. There is a lower likelihood of having an IQ [30].

A person's psychological attachment and loyalty to an organization are referred to as organizational commitment [31]. This study emphasizes affective commitment. It impacts an employee's behaviour, including their decision to remain with the company. It has an adverse relationship with the intent of quitting. Because emotionally involved individuals are more likely to be engaged, motivated, and satisfied with their work, organizations consistently desire to encourage higher levels of affective commitment [32].

In this study, job satisfaction refers to a person's overall contentment, happiness, and good emotional state as a result of their work experiences [33]. Lower turnover among employees is associated with high job satisfaction. Employees are more likely to stick in the company for an extended period of time when they are happy with their jobs. This minimizes the cost of hiring, preparing, and integrating new staff. On the contrary, job dissatisfaction increases the likelihood of intending to quit [34]. Despite the recent literature on the many antecedents of employee intention to quit, research into the relationship between job involvement, organizational commitment, job satisfaction, and employee intention to quit is still ongoing and must be focused on and emphasized [35,36].

2.1.2. Activities in professional networking sites in social media (APNS), update of profile in job portals (PJP) and Intention to quit (IQ)

Social networking, which is an emerging field for employees, can be beneficial. It complements technology to encourage employee comprehension, collaboration, and relationship building. Several studies have investigated how businesses may use social media to engage their employees and immerse them in the game, resulting in higher performance and lower turnover [11,37,38]. As a result of increased workplace integration, there is a lower likelihood of turnover [39]. However, frequent use of social media for non-work-related purposes or in leisure time raises the chance of employee turnover [38]. Furthermore, this makes employees more likely to engage in behaviours associated with job searching, which could lead to their exit from their current workplace. Unveiling the model of voluntary turnover and incorporating social media presence demonstrates that professional development on networking sites influences an individual's job search behaviour and causes them to leave the company [18]. As a result, activity on professional networking sites and updating profiles on employment portals on social media are variables in predicting employee IQ [19].

Previous studies have primarily focused on examining individual predictors of IQ, such as job satisfaction, job involvement, organizational commitment, activities on professional networking sites, and updates of profiles in job portals, in isolation. These studies have contributed valuable insights into how these factors independently influence employees' likelihood to quit their jobs. However, a crucial gap arises from the lack of exploration into how these predictors interact and collectively contribute to employees' decision-making processes regarding their intention to quit. The current study addresses this gap by establishing the relationship between predictors or features such as Job Satisfaction, Job Involvement, Organizational Commitment, Activities on professional networking sites, updates of Profiles in job portals, and introducing the Stimulus–Organism–Response (S-O-R) theory as a framework to predict intention to quit. This theory offers a comprehensive lens through which external stimuli (job-related factors) influence employees' internal states (emotions, attitudes), ultimately leading to their behavioural response of intent to quit or stay.

2.2. Ensemble techniques

The turnover prediction approach employing Ensemble learning, as demonstrated in recent empirical work, simplifies a standard supervised binary classification issue [40]. In the Machine Learning process, the Ensemble receives a lot of attention and focus. It displays an exceptional ability to improve categorization accuracy. The ensemble technique, which combines many classifiers, significantly enhanced the learning algorithm's prediction performance with classifiers including KNN, bagged KNN, random KNN, multiple feature subset method, random forest, and support vector machines [41]. Qutub et al. [42] use ensemble learning with a Decision tree, logistic regression, and SVM for classifying the IQ on the IBM HR analytics data and the ensemble learning model outperforms all three models. The performance of SVM ensembles outperforms that of other approaches with a higher degree of generality [43]. The suggested ensemble technique focuses on a weighted classification model based on individual projected accuracy. Based on a weighted average, the personnel can be divided into two groups: IQ and Non-IQ. Organizations will be able to take the required steps towards controlling turnover with the support of precise classification predictions [44].

However, there is a dearth of studies that have taken a holistic approach by leveraging classification tools. The application of classification algorithms such as decision trees, Gradient Boosting, Random Forests, and KNN has the potential to uncover intricate interactions and nonlinear relationships among the various predictors, leading to more precise predictions of an employee's intention to quit. Therefore, the integration of the Stimulus–Response theory and the use of a Classification tool together contribute to the research gap between previous research and the current study. By creating a comprehensive model that incorporates the Stimulus Organism Response theory, the study sheds light on the prediction of the employees' intentions to quit. Therefore, the aim of this study is to develop an ensemble learning classification model for predicting IQ by leveraging the established relationships among the predictors, showcasing the optimal combination of methods to build a classification model, and integrating the stimulus organism response theory to enhance the predictive framework. If this prediction technology is developed successfully, it will give organizations a useful way to proactively identify at-risk employees, enabling them to undertake focused retention measures and eventually improving employee happiness and organizational stability.

3. Methodology and data

3.1. Experiment methods — data set impression and collection procedure

The empirical data were gathered using cross-sectional survey techniques. The Information Technology (IT) personnel who are exposed to the IoT were the participants in this study. An online questionnaire using SurveyMonkey was distributed among the employees of selected IT organizations. The data were collected using simple random sampling. Randomly the online questionnaire was distributed to around 1006 employees of a well-known IT organization and only Six hundred and twenty-one (621) participants responded. Twenty-four responses were incomplete and had to be discarded. This left 597 valid responses for the statistical analysis. In this collected 597 data, the shares of male and female respondents were 63% and 37%, respectively. The average age of the respondents was 35 years, and practically all of them had an average of 11 years and 4 months of experience. A total of 35 questions were asked to these 597 respondents and these questions are indicators. The data was collected during the period between 2021–2022. The measures of the variables are provided in Table 1.

The responses are collected using 1 to 5 Likert scale and summary of the collected data is provided in Table 2 as descriptive statistics.

Table 1
Measures of variables.

S.No	Variable	Items/ Indicators	Author
1	Job Involvement (JI)	5	Kanungo's (1982) and Lodahl and Kejnar [29].
2	Organizational Commitment (OC)	6	Mowday et al. [45]
3	Job Satisfaction (JS)	9	Warr et al. [46]
4	Activities in Professional Networking Sites (APNS)	8	Biswas et al. [19]
5	Profile in Job Portal (PJP)	3	Biswas et al. [19]
6	Intention to Quit (IQ)	4	Hom and Griffeth [47], Sjöberg and Sverke [48].

Table 2
Descriptive statistics of the input variables.

Variables		Descriptive statistics				
		N	Minimum	Maximum	Mean	Std. deviation
Work Perfectionism	JI1	597	1	5	3.63	1.254
Job Existentialism	JI2	597	1	5	3.49	1.115
Workaholic/Job Involvement	JI3	597	1	5	3.28	1.076
Personal Development	JI4	597	1	5	3.19	1.091
Job Dedication	JI5	597	1	5	3.01	1.008
Outperform	OC1	597	1	5	3.77	1.063
Role acceptance	OC2	597	1	5	3.50	1.029
Organizational proud	OC3	597	1	5	3.65	1.068
Loyalty	OC4	597	1	5	3.72	1.058
Value homogeneity	OC5	597	1	5	3.61	1.092
Business Process	OC6	597	1	5	3.61	.957
Working environment	JS1	597	1	5	3.39	1.193
Compensation & Benefits	JS2	597	1	5	3.34	1.165
Sense of accomplishment	JS3	597	1	5	3.54	1.062
Initiatives	JS4	597	1	5	3.38	1.007
Micro management	JS5	597	1	5	3.70	1.150
Career progression	JS6	597	1	5	3.39	1.058
Team engagement	JS7	597	1	5	3.83	1.018
Professional recognition	JS8	597	1	5	3.48	1.059
Management Support	JS9	597	1	5	3.47	1.011
Social Media Usage Frequency	APNS1	597	1	5	3.77	1.080
Social Media Preferences	APNS2	597	1	5	3.70	1.018
Social Media Preferred Activities	APNS3	597	1	5	3.49	.967
Profile updation	APNS4	597	1	5	3.82	.969
Subscriptions	APNS5	597	1	5	3.52	1.039
Membership type	APNS6	597	1	5	3.57	1.053
Time spent	APNS7	597	1	5	3.04	1.056
Job search preference	APNS8	597	1	5	3.45	1.111
Applies to Job Advertising	PJP1	597	1	5	3.36	1.194
Updating CV for Recruiters call	PJP2	597	1	5	3.33	1.064
Updating CV for Current Opening	PJP3	597	1	5	3.04	1.054
Intention to Quit	IQ	597	0	1	0.956	0.204

3.2. Questionnaire design

The questionnaire had two main sections: one for the respondent's basic information and the other for their responses to our study constructs. The subject's demographic data (such as gender, age, designation, and experience, past and present) were entered in the basic data section. The participant's impressions on each model variable were captured in the second section. Each construct's associated items are included. A 5-point scale with a range of 1 (strongly disagree) to 5 (strongly agree) is used to evaluate each item.

The output or target variable for the study, Intention to Quit (IQ), was obtained from the same questionnaire in the form of a yes or no response and further translated into a numerical value of 1 or 0, which can be seen in Table 2 along with the mean and standard deviation.

The questionnaire's primary use is to gather information from any organization and use it as input into the classification model to determine the intention to quit. Building a classification model for predicting an employee's intention to quit (IQ) is the primary goal of the study, as was already mentioned. To determine whether employees would or might not quit their jobs, one can use this classification model as a tool. The classification logic is described using an ensemble of six well-known classification algorithms, including Naive Bayes, SVM, Random Forest, KNN, Logistic Regression and Neural Network. The IQ is predicted using the proposed ensemble learning-based methodology. For the purpose of proving its superiority, this model is also contrasted with the six distinct standalone classifiers employed in ensemble learning. The following subsections provide descriptions of the individual models.

3.3. Feature information gain analysis

In this study, information gain is estimated in order to understand the significance and status of entropy in the variable. Evidence of reduced uncertainty in that variable or feature include lower entropy, which is demonstrated by higher information gain. The calculated information gain is provided in Table 3 based on Eq. (1) [49]. Table 3

Table 3
Calculated information gain for feature variables.

Factor/Feature	Information gain	Rank	Factor/Feature	Information gain	Rank
JI4	0.0183	1	JI1	0.0066	15
OC4	0.0179	2	APNS2	0.0065	16
APNS6	0.0176	3	JS5	0.0058	17
OC3	0.0174	4	JS9	0.004	18
JS8	0.0156	5	JS3	0.0034	19
JI3	0.0149	6	APNS4	0.0029	20
PJP2	0.0146	7	OC6	0.0026	21
APNS8	0.0128	8	JS1	0.0021	22
JI2	0.0128	9	JS6	0.0021	23
APNS5	0.0112	10	PJP3	0.0019	24
APNS1	0.0111	11	OC1	0.0019	25
PJP1	0.0097	12	OC5	0.0011	26
APNS3	0.0089	13	OC2	0.0009	27
JS2	0.0088	14			

Table 4
Correlation coefficient values.

	JI1	JI2	JI3	OC2	OC3	JS3	APNS1	APNS2	APNS4	APNS5	APNS6	PJP1
JI2	**.564****											
JI3		**.508****										
JI4			**.526****									
OC3				**.557****								
OC4				**.514****	**.763****							
JS4						**.585****						
APNS2							**.540****					
APNS4							**.597****	**.525****		.		
APNS6									**.506****	**.548****		
APNS8											**.594****	
PJP2												**.584****

Correlation value: $(0 \leq \eth r_p \eth < 0{:}3)$ *low correlation value, Correlation value:* $(0{:}3 \leq \eth r_p \eth < 0{:}6)$ *moderate correlation, Correlation value:* (< 1) *: high correlation value.*
** *Correlation is significant at the 0.01 level (2-tailed).*

data indicates JI4 have highest value of information gain followed by OC4, APNS5, OC3, JS8, and JI3.

$$IG = -\sum_{i=1} p\ c_i\ \log p\ c_i\ + p(t) \sum_{i=1} p\ c_i \eth t\ \log p\ c_i \eth t + p\ \bar{t}\ \sum_{i=1} p\ c_i \eth \bar{t}\ \log p\ c_i \eth \bar{t} \quad (1)$$

Where, c_i represents the ith category, $p\ c_i$ is the probability of the ith category; $p(t)$ and $p\ \bar{t}$ are the probabilities that the term t appears or not in the data set; $p\ c_i \eth t$ is the conditional probability of the ith category given that term t appeared, and $p\ c_i \eth \bar{t}$ is the conditional probability of the ith category given that term t does not appear.

3.4. Feature correlation matrix

The correlation analysis has been conducted to identify the association between the features used in building the classification model and Pearson correlation calculation method is deployed for this research [50]. The expression for the correlation calculation is provided in Eq. (2). The calculated Pearson correlation coefficient values are provided in Table 4.

$$r_p = \frac{\sum_{i=1}^{N} x_i - \overline{x_i}\ \ y_i - \overline{y_i}}{\sqrt{\sum_{i=1}^{N} x_i - \overline{x_i}^{\,2}}\ \sqrt{\sum_{i=1}^{N} y_i - \overline{y_i}^{\,2}}} \quad (2)$$

Where, r_p is Pearson correlation coefficient. x_i and y_i are two variables. $\overline{x_i}$ and $\overline{y_i}$ are the averages of variable. For the purpose of explanation, the correlation values are broadly divided into three categories one is low $(0 \leq \eth r_p \eth < 0{:}3)$, moderate $(0{:}3 \leq \eth r_p \eth < 0{:}6)$ and another is high $(1 \geq \eth r_p \eth \geq 0{:}6)$.

Based on the calculated value, the highly correlated feature pairs are (JI1, JI2), (JI2, JI3), (JI3, JI4), (OC2, OC3), (OC2, OC4), (OC3, OC4). (JS3, JS4), (APNS1, APNS2), (APNS2, APNS4), (APNS4, APNS6), and (APNS5, APNS6). Each of these feature pair correlations has a higher significance level of 99%, indicating the significance of the linkages in relation to the data that was collected. The result discussion section includes a detailed description of the explanation for these relationships. The feature pairings that are not the pairs with lower correlation coefficient values show a weaker relationship between them.

3.5. IQ prediction tool modelling

The proposed tool for predicting the IQ is modelled by combining five well known classification algorithm. These five classification algorithms are Gradient Boosting, Random Forest, KNN, SVM and Naive-Bayes. This ensemble model is trained with 70% of the total collected data and remaining 30% is used for testing of the model. The methodological flow of this entire study has been presented in Fig. 1.

3.5.1. Training data preparation

This study's primary goal is to model IQ, and in order to do that, an accurate prediction is needed. Therefore, in order to ensure that the training data is accurate or devoid of skewness towards a specific class, 209 data points from the positive sample and another 209 data points from the negative sample, totalling 70% of the 418 data points, are randomly selected. This guarantees that there is no bias towards any single class and greatly increases the tool's robustness. The 179 remaining data points are considered for testing the proposed tool or model.

3.5.2. Algorithms

The major goal of the ensemble technique is to increase the proposed tool's capacity for prediction by utilizing a variety of classification algorithms.

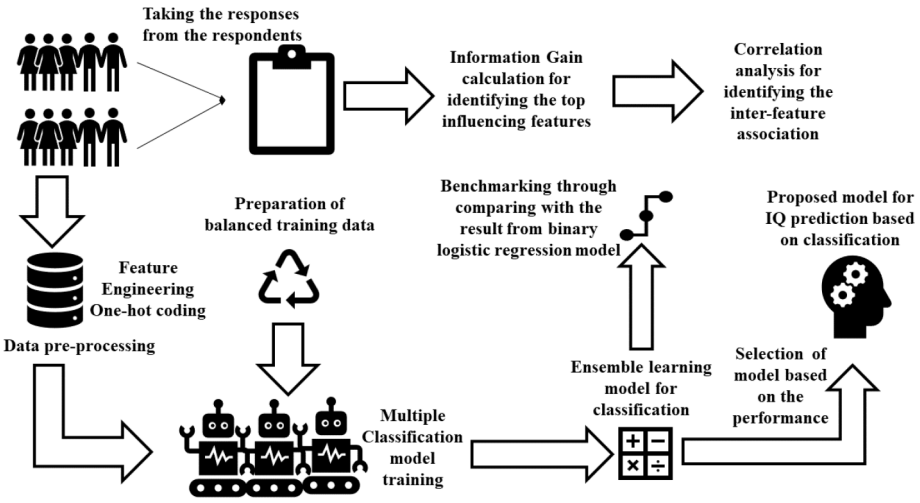

Fig. 1. Methodological flow.

Table 5
Gradient boosting algorithm.

Step	Description
1	Initialize input data $(x;y)_{i=1}^{N}$, number of iteration M, loss-function $L(y; f)$, base learning model $h(x;)$, Initialize f_0 with a constant

2	For t=1 to M do
	Compute the negative gradient $g_t(x)$
	fit a new base-learner function $h(x; _t)$
	find best gradient descent step-size $_t$
	$_t = \text{argmin} \sum_{i=1}^{N} [y_i; f_{t-1}\ x_i + h\ x_i; _t]$
	Update the function estimate $f_t \leftarrow f_{t-1} + _t h\ x_i; _t$
	End For

Table 6
Random Forest algorithm.

Step	Description
1	Initialize N is quantitative number of samples, M indicates total number of features, m is sample size, k indicates next node
2	Creates N samples from the dataset
3	Every node taking a feature randomly of size m where m<M.
4.	Builds a split for the m features selected in step 2 and detects the k node by using the best split point.
5.	Split the tree iteratively until one leaf node is attained and the tree remains completed.
6.	The algorithm is trained on each bootstrapped independently.
7.	Using trees classification voting predicted data is collected from the trained trees.
8.	The final RF model is build using the peak voted features.
9	End

3.5.2.1. Gradient Boosting. Gradient Boosting is a machine learning model that tries to build a strong classification model through combining weak learner models. The strong model is built by reducing the value of loss function or cross-entropy during the training process. This minimization of loss function is achieved through gradient decent method. The algorithm has been provided here stepwise in Table 5.

Gradient Boosting, the first algorithm utilized, has several advantages over other existing techniques. Better class prediction accuracy is the first benefit, and shorter training times is the second. Thanks to these, it can handle massive data sets, but the key issue is that the model is difficult to interpret [51]. This technique is used in other disciplines fields such as psychology [52] and consumer behaviour Arghir [53], this technique is applied.

3.5.2.2. Random Forest. The second method under examination is Random Forest, which primarily benefits from considering various decision trees designed to represent the rules for classification [54]. According to Schmitt [55], Random Forest is one of the prevalent techniques for creating classification models. The algorithm has been provided in Table 6.

3.5.2.3. K Nearest Neighbour. The other K Nearest Neighbour (KNN) technique is effective at handling multiclass models and can also maintain the data's nonlinearity. Additionally, it can learn with fewer data (Cheng, 2014). According to neighbourhood data, the KNN is already a tried-and-true classification technique [56,57]. The algorithm has been provided in Table 7.

3.5.2.4. Support vector machine. Although support vectors are an excellent tool for classification that can handle a large number of features and maintain nonlinearity during training, they are not always effective [58]. The algorithm has been provided is Table 8.

Table 7
KNN algorithm.

Step	Description
1	Initialize input data $(x;y)_{i=1}^{N}$, number of nearest neighbour k (odd number), loss-function d(x; y) ,test sample x
2	For each training sample $x_i; y_i \in D$ do Compute d(x, x_i), the distance between x and x_i Let N Ó D be the set of training samples with the k smallest distance d(x, x_i) Return the majority label of the samples in N

Table 8
Support vector machine algorithm.

Step	Description
1	Initialize the input data $(x;y)_{i=1}^{N}$

2	Determine the calculated accuracy
	Select the optimal value of cost and gamma value for support vector machine.
	While (stopping condition is not met) do
	Implement SVM train step for each data point.
	Implement SVM classify for testing data points.
	End while
	Return accuracy

Table 9
Naive Bayes classifier.

Step	Description
1	Initialize the input data $(x;y)_{i=1}^{N}$
2	Calculate the mean and standard deviation of the predictor variable in each class Repeat Calculate the probability of predicted value using the density equation in each class Until the probability of all predictor features has been calculated
3.	Calculate the likelihood for each class and find out the highest likelihood value.

Table 10
Neural network classifier algorithm.

Step	Description
1	Initialize the input data $(x;y)_{i=1}^{N}$, initialize the weights w_i, he bias b_i, activation function f, is learning rate.
2	Train the model based on the relation For i=1 to N $\hat{y}_i = f(w_i x_i + b_i)$ $w_{new} = w_{old} - \frac{de}{dw}$ $b_{new} = b_{old} - \frac{de}{db}$ Iterate until error e is lowest. End

3.5.2.5. Naive Bayes classifier. However, compared to other classification algorithms, Naive Bayes is a basic technique that can handle a variety of data. In the many research publications published in the human resource management area [59–61], this algorithm has been implemented. The algorithm has been provided in Table 9.

3.5.2.6. Neural network. An Artificial neural network is a computational model inspired by the biological neural networks of the human brain. Neural Networks are designed to learn and adopt from data, enabling them to recognize patterns, make prediction and solve complex problems. The use of neural network can be found in Human Resource Management in the article by Tung et al. [62]. The algorithm has been provided in Table 10.

3.5.2.7. Logistic regression. Logistic regression is a statistical modelling technique used to predict the probability of a binary outcome based on a set of variables [63]. It is commonly employed in machine learning and statistical analysis for classification tasks. Application of logistic regression in HRM can be found in [64,65]. The algorithm has been described in Table 11.

Table 11
Logistic regression classification algorithm.

Step	Description
1	Initialize the input data $(x;y)_{i=1}^{N}$
2	Train the model based on the input data For i=1 to N di=(xi,yi) set the target value for the regression to $Z_i = \frac{y_i - p(1\partial d_j)}{[p(1\partial d_j)(1-p(1\partial d_j))]}$ Initialize the weight of instance d_j to $p\ 1\partial d_j\ \ 1-p\ 1\partial d_j$ Finalize a f(j) to the data with class value (Zj) and weight (wj) If pi>0.5 Assign class label 1 Otherwise Assign class label 2 End End

3.5.2.8. Ensemble learning. A technique known as the ensemble technique is used to combine the multiple classification algorithms previously stated. This increases the model's accuracy and robustness while decreasing prediction error [66]. In this article, the stacking type of ensemble model is used due to the several advantages as mentioned in published articles [67–69]. Stacking or blending permits more flexibility in combining the base models compared to other ensemble learning approaches. It uses a metamodel-based approach, which identifies the

Fig. 2. Process for the stacking method.

Table 12
Confusion matrix structure.

		Predicted	
		1	0
Actual	1	True Positive (TP)	False Negative (FN)
	0	False Positive (FP)	True Negative (TN)

pattern in the prediction of base models and learn the best possible way to give the best prediction. This approach is different than simple average or majority voting. The stacking can handle the individual base model prediction error and based on this down weight the model prediction with the highest error. This also increases the bias towards the base models who have higher prediction accuracy. Stacking allows different kinds of base models in ensemble process, such as decision tree and neural network, which helps to exploit the complementary properties of such models. As a result of this, stacking, produces prediction based on the strengths and weaknesses of base models, which leads to higher prediction accuracy. This stacking approach is represented in Fig. 2. In this figure, it can be observed first the base models are trained using the data and the output from the base models are fed to another base model which works as meta model. Here the output from the base models is act as features for the meta model.

3.6. Performance measurement and comparison

3.6.1. Confusion matrix

Confusion matrix is a summary of right and wrong predictions of both positive and negative class. In Table 12, a simple example of confusion matrix is provided. The number of rightly predicted positive case is called True Positive (TP) and the number of wrong predictions for positive classes are known as False Positive (FP). On the other hand, the number of right predictions for negative class is True Negative (TN) and number of wrongly predicted negative class is known as False Negative (FN). These numbers are then used to calculate the models' performance metrics. In this study, Classification Accuracy (CA), F1 Score, Precision and Recall are calculated [70]. The calculated confusion matrix for all the models has been provided in Table 13 for the test dataset. A1 and A0 are indicating the actual positive and negative classes where P1 and P0 are indicating the predicted positive and negative classes respectively.

3.6.1.1. Accuracy (CA). The classification accuracy can be measure by taking ratio of total number of right prediction and total number of predictions. The numerical expression for the accuracy is provided in Eq. (3).

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{3}$$

3.6.1.2. F1 score. F1 Score considers both precision and recalls to compute the score. The F1 score can be interpreted as a weighted average of the precision and recall values, where an F1 score reaches its best value at 1 and worst value at 0. The expression for calculation is presented in Eq. (4).

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

3.6.1.3. Precision. Precision is measured by calculating the ratio of detected positive cases and total predicted positive cases. The precision can be computed based on Eq. (5).

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

3.6.1.4. Recall. Recall is the ratio of correctly predicted positive cases and total available positive case inside a dataset. The recall can be computed based on Eq. (6).

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

3.6.1.5. Receiver Operating Characteristic Curve (ROC). A ROC Curve is a graph showing the performance of a classification model at all classification thresholder's TP ratio values are plotted with respect to FP ratio.

4. Result discussion

4.1. Classification of IQ

With the advent of the modern era, demonstrating a significant change in the workplace has become a competitive advantage for the workforce. Cloud-based working and collaboration with social media networking are two examples of innovations that require more HR attention. The desire of the workforce to work in cutting-edge technology influences job selection among available positions. Exposure to social media as a technology investment may provide a diverse range of egalitarian work settings for the workforce. Therefore, the positive and negative evaluation of the usage of social media in the workplace demands a great attention of the HR Managers in an organization. The use of social media in non-work-related context holds the working environment. It changes the attitude of an individual employee manifesting their withdrawal behaviour which affects the bottom line of the organization's performance. Therefore, from the standpoint of an organization, it is critical to empower individual employees to attract and retain talent.

As a result, the classification of intention to quit acts as a benchmark for improving talent retention decision-making. In this study, Intention to Quit is considered as a class label based on the various predictors. When a person is assigned to the "Yes" category, it means that they are